

SCALAR REGULAR POLYNOMIAL LATTICES UNDER DUAL LATTICE ISOMORPHISM

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STATUS

This packet gives a likely valid full scalar answer to the question highlighted in Boyd–Miranda [1]. The source asks, in lattice form, when a lattice isomorphism $E^* \simeq F^*$ forces the scalar polynomial lattices $\mathcal{P}^r({}^n E)$ and $\mathcal{P}^r({}^n F)$ to be lattice isomorphic. The source proves a stronger vector-valued theorem under the additional hypothesis that one dual has order-continuous norm. The theorem below removes that hypothesis for the scalar-valued regular-polynomial question.

SOURCE PASSAGE

The introduction of [1] asks:

Given two dual isomorphic Banach lattices E and F , that is E^* and F^* are lattice isomorphic, under which conditions are $\mathcal{P}^r({}^n E)$ and $\mathcal{P}^r({}^n F)$ lattice isomorphic for every $n \in \mathbb{N}$?

The same introduction states the source paper’s sufficient hypothesis: if E^* and F^* are lattice isomorphic and have order-continuous norms, then $\mathcal{P}^r({}^n E; G^*)$ and $\mathcal{P}^r({}^n F; G^*)$ are lattice isomorphic for every Banach lattice G .

1. Zaidenando in [29] who also addressed the same question concerning other classes of polynomials. Further, in [17], D. Carando and S. Lassalle extended these results to spaces of vector-valued homogeneous polynomials.

Motivated by the increasing interest in studying homogeneous polynomials on Banach lattices (see, e.g., [6, 7, 8, 9, 10, 11, 13, 24, 31]), the following question arises naturally: *Given two dual isomorphic Banach lattices E and F , that is E^* and F^* are lattice isomorphic, under which conditions are $\mathcal{P}^r({}^n E)$ and $\mathcal{P}^r({}^n F)$ lattice isomorphic for every $n \in \mathbb{N}$?* By adapting the proofs from [17] and [29], we were able to provide a positive answer whenever E^* and F^* have order continuous norms. In particular, we prove that if E^* and F^* are lattice isomorphic with at least one (and hence the other) having order continuous norm, then $\mathcal{P}^r({}^n E; G^*)$ and $\mathcal{P}^r({}^n F; G^*)$ are lattice isomorphic for every $n \in \mathbb{N}$ and every Banach lattice G . We observe that, in our setting, the assumption of Arens regularity is replaced

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DEFINITIONS

All Banach lattices are real. For a Banach lattice E , let $J_E : E \rightarrow E^{**}$ be the canonical lattice embedding and set

$$E^\diamond := (E^*)^*_n,$$

the band of order-continuous linear functionals on the Banach lattice E^* . The canonical copy $J_E(E)$ is contained in $E^\diamond$. For a regular n -homogeneous scalar polynomial P on E , write $\check{P}$ for its associated symmetric regular n -linear form.

IDEA OF THE PROOF

The source uses Arens extensions on the whole bidual, so it needs $E^{**} = E^\diamond$. For the scalar problem, the whole bidual is unnecessary. By the order-continuity theorem of Botelho–Garcia [2], every Arens extension of a scalar regular multilinear form is separately order continuous on the product of the order-continuous bidual bands. Since the canonical image of E is order dense in $E^\diamond$ in the Fremlin order-bidual sense, any two such separately order-continuous extensions that agree on E^n agree on $(E^\diamond)^n$. Thus a symmetric regular form has a canonical symmetric extension to $(E^\diamond)^n$.

If $\varphi : E^* \rightarrow F^*$ is a lattice isomorphism, then $\varphi^* : F^{**} \rightarrow E^{**}$ maps $F^\diamond$ onto $E^\diamond$. A polynomial P , with associated form $A = \check{P}$, can therefore be transferred by

$$\Psi_\varphi(P)(y) = A^\diamond(\varphi^* J_F y, \dots, \varphi^* J_F y), \quad y \in F.$$

The same order-density argument gives the functorial identity needed to see that $\Psi_{\varphi^{-1}}$ is the inverse of Ψ_φ . Positivity is preserved by positivity of Arens extensions of positive multilinear maps.

FULL SCALAR THEOREM

Theorem 1. *Let E and F be Banach lattices. If E^* and F^* are lattice isomorphic, then, for every $n \in \mathbb{N}$, the Banach lattices*

$$\mathcal{P}^r({}^n E) \quad \text{and} \quad \mathcal{P}^r({}^n F)$$

are lattice isomorphic.

Lemma 2. *Let $A : E_1 \times \dots \times E_n \rightarrow \mathbb{R}$ be a regular n -linear form on Banach lattices. Then all Arens extensions of A have the same restriction to*

$$E_1^\diamond \times \dots \times E_n^\diamond.$$

This common restriction, denoted $A^\diamond$, is separately order continuous. If $E_1 = \dots = E_n = E$ and A is symmetric, then $A^\diamond$ is symmetric. If A is positive, then $A^\diamond$ is positive.

Proof. For every Arens extension and every permutation, separate order continuity on $E_1^\diamond \times \cdots \times E_n^\diamond$ is exactly the order-continuous-bidual part of the main Section 4 theorem of [2], which improves the Boyd–Ryan–Snigireva and Buskes–Roberts Arens-extension theorems recalled in that paper.

The extensions agree on $J_{E_1}(E_1) \times \cdots \times J_{E_n}(E_n)$. The Fremlin order-bidual construction gives

$$DIDI J_E(E) = E^\diamond$$

for each Banach lattice E ; see the DIDI identity used in [3]. If two regular n -linear forms are separately order continuous on the product of the $E_i^\diamond$ and agree on the canonical product, equality propagates through the I , D , I , and D order closures one coordinate at a time. Hence all Arens extensions agree on the displayed product.

If A is symmetric, applying the preceding uniqueness to the permuted extensions shows that the common restriction is invariant under coordinate permutations. If A is positive, positivity follows from positivity of Arens extensions of positive multilinear maps, as used in [2, Theorem 2.2(c)]. $\square$

Lemma 3. *Let $\varphi : E^* \rightarrow F^*$ be a lattice isomorphism. For $P \in \mathcal{P}^r({}^nE)$ with associated form $A = \check{P}$, define*

$$\Psi_\varphi(P)(y) = A^\diamond(\varphi^* J_F y, \dots, \varphi^* J_F y), \quad y \in F.$$

Then $\Psi_\varphi(P) \in \mathcal{P}^r({}^nF)$, and the associated symmetric form $B = \Psi_\varphi(\check{P})$ satisfies, for all $\zeta_1, \dots, \zeta_n \in F^\diamond$,

$$(1) \quad B^\diamond(\zeta_1, \dots, \zeta_n) = A^\diamond(\varphi^* \zeta_1, \dots, \varphi^* \zeta_n).$$

Moreover, Ψ_φ is positive.

Proof. First φ^* maps $F^\diamond$ onto $E^\diamond$. Indeed, if ζ is order continuous on F^* and $x_\alpha \xrightarrow{o} 0$ in E^* , then $\varphi x_\alpha \xrightarrow{o} 0$ in F^* because lattice isomorphisms preserve order convergence. Hence $(\varphi^* \zeta)(x_\alpha) = \zeta(\varphi x_\alpha) \rightarrow 0$. The converse follows from φ^{-1} .

For $y_1, \dots, y_n \in F$ set

$$B(y_1, \dots, y_n) = A^\diamond(\varphi^* J_F y_1, \dots, \varphi^* J_F y_n).$$

By Lemma 2, $A^\diamond$ is symmetric whenever A is symmetric, so B is a symmetric regular n -linear form. Its diagonal is $\Psi_\varphi(P)$, hence $\Psi_\varphi(P)$ is a scalar regular n -homogeneous polynomial. If $P \geq 0$, then $A \geq 0$, so $A^\diamond \geq 0$ and $B \geq 0$; therefore $\Psi_\varphi(P) \geq 0$.

To prove (1), compare two regular n -linear forms on $(F^\diamond)^n$:

$$H(\zeta_1, \dots, \zeta_n) = B^\diamond(\zeta_1, \dots, \zeta_n)$$

and

$$K(\zeta_1, \dots, \zeta_n) = A^\diamond(\varphi^* \zeta_1, \dots, \varphi^* \zeta_n).$$

Both are separately order continuous: H by Lemma 2, and K because $A^\diamond$ is separately order continuous and φ^* is a lattice isomorphism from $F^\diamond$ onto

$E^\diamond$. They agree on $J_F(F)^n$ by the definition of B . The same Fremlin order-density argument used in Lemma 2 therefore gives $H = K$ on $(F^\diamond)^n$. $\square$

Proof of Theorem 1. Fix a lattice isomorphism $\varphi : E^* \rightarrow F^*$ and set $\theta = \varphi^{-1}$. Lemma 3 gives positive linear maps

$$\Psi_\varphi : \mathcal{P}^r({}^n E) \rightarrow \mathcal{P}^r({}^n F), \quad \Psi_\theta : \mathcal{P}^r({}^n F) \rightarrow \mathcal{P}^r({}^n E).$$

Let $P \in \mathcal{P}^r({}^n E)$, let $A = \check{P}$, and let $B = \Psi_\varphi(\check{P})$. For $x \in E$, apply (1) to $\theta^* J_E x, \dots, \theta^* J_E x \in F^\diamond$:

$$\begin{aligned} (\Psi_\theta \Psi_\varphi P)(x) &= B^\diamond(\theta^* J_E x, \dots, \theta^* J_E x) \\ &= A^\diamond(\varphi^* \theta^* J_E x, \dots, \varphi^* \theta^* J_E x) \\ &= A^\diamond(J_E x, \dots, J_E x) = P(x). \end{aligned}$$

Thus $\Psi_\theta \Psi_\varphi = I$ on $\mathcal{P}^r({}^n E)$. The symmetric calculation gives $\Psi_\varphi \Psi_\theta = I$ on $\mathcal{P}^r({}^n F)$. Hence Ψ_φ is a positive bijection whose inverse is positive, so it is a lattice isomorphism of Banach lattices. $\square$

VERIFICATION NOTES AND LIMITATIONS

The proof is analytic. No numerical check is relevant. The bounded local novelty check searched the run indexes for the arXiv id, the exact title, dual lattice isomorphism, scalar regular polynomials, orthogonally additive polynomials, and order-continuous bidual phrases. The only duplicate was the earlier orthogonally additive scalar partial packet created by this lane, which this packet upgrades. A bounded web search on 2026-06-29 for exact and near-exact formulations of the scalar upgrade surfaced the source paper but no later exact answer.

This settles the extracted scalar question. It does not settle the vector-valued G^* -valued theorem without the source paper's order-continuity assumptions, and it does not address arbitrary Banach-space dual isomorphisms that are not lattice isomorphisms.

REFERENCES

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